Model Choice Matters More Than Prompts: Evaluating LLM Story Judges Across Techniques and families

Nouf AlMansour **Nour AlGhomlas** **Tarfah Bin Moammar** **Shahad AlMutairi** **Lujain AlHarbi**
King Saud University
College of Computer and Information Sciences
Department of Information Technology
{444200525, 444200611, 444200935, 444200811, 443200811}@student.ksu.edu.sa
Supervised by: Hend Alrasheed

Abstract

Large language models (LLMs) are increasingly used as automatic evaluators of generated text, but their reliability for subjective tasks such as story evaluation depends heavily on how they are prompted. There is no consensus on which prompting strategy yields the most human-aligned scores, and little controlled comparison across model families. In this paper, we systematically compare five prompting strategies for LLM-based story evaluation. We apply each strategy to GPT-4.1 Mini and one of the Claude variants (Sonnet 4.5, Haiku 4.5) on the HANNA benchmark, producing over ten thousand evaluations across six narrative dimensions, and compare the resulting scores against human annotations. Our analysis shows that simpler prompts outperform more elaborate personas, that model choice matters more than prompt design, and that LLM judges rank stories reliably but produce poorly calibrated absolute scores—findings that motivate practical recommendations for using LLMs as story evaluators.

1 Introduction

Large language models (LLMs) have demonstrated strong capabilities in generating coherent, structured, and contextually rich narratives (Brown et al., 2020; OpenAI, 2023). Recent systems can produce short stories that exhibit grammatical fluency, logical progression, and creative elements such as plot twists, emotional arcs, and stylistic variation. As these models become increasingly integrated into creative writing tools, recommendation systems, and automated content generation pipelines, a critical question emerges: **can LLMs reliably evaluate the quality of the stories they generate?**

1.1 Motivation

Story quality is inherently multidimensional and subjective. Human evaluation typically considers aspects such as relevance to the prompt, narrative coherence, emotional expressiveness (e.g., empathy), creativity (e.g., surprise), engagement, and linguistic complexity. Traditionally, such evaluations are performed by human annotators, making the process expensive, time-consuming, and difficult to scale.

Consequently, there has been growing interest in using LLMs as automatic evaluators of generated text. If LLMs can reliably approximate human judgment, they could significantly reduce annotation costs and enable scalable evaluation of generative systems. However, a central concern is whether different prompting strategies and model architectures produce consistent, reliable evaluations that align with human judgment across diverse narrative dimensions.

1.2 Brief State of the Art

Prior work in automatic story generation (ASG) evaluation has explored both traditional and neural-based approaches. Early methods relied on surface-level metrics such as BLEU and ROUGE (Papineni et al., 2002; Lin, 2004), which capture lexical overlap but fail to reflect narrative quality or creativity. More recent approaches introduce learned evaluation models and embedding-based metrics such as BERTScore (Zhang et al., 2020).

With the emergence of LLMs, a new paradigm has gained traction: **LLM-as-a-judge**, where large language models are directly prompted to evaluate generated content (Zheng et al., 2023; Chiang et al., 2023). While these systems can approximate human rankings under carefully designed prompts, prior work highlights limitations such as prompt sensitivity, instability, and bias toward fluent outputs.

Despite these advances, relatively little work has systematically compared multiple prompting strategies across different LLM families using a consistent human-annotated benchmark for narrative evaluation.

1.3 Research Goal

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> **How do different prompting techniques affect LLM story evaluation compared to human judgments?**

To address this, we systematically compare five prompting strategies (role-based personas of varying complexity, procedural instructions, and zero-shot baseline) across two model families (GPT-4.1 Mini and Claude variants) on 1,056 human-annotated stories, analyzing alignment using error metrics, agreement measures, and correlation coefficients.

1.5 Main Contributions

This work makes the following contributions:

1. A comprehensive evaluation framework comparing LLM-based story evaluations against human annotations across six narrative dimensions.

2. A multi-model comparison involving GPT-4.1 Mini and Claude variants (Haiku 3, Sonnet 4.5, Haiku 4.5).

3. A prompt-level analysis across five prompting strategies.

4. A calibration analysis examining the gap between ranking ability and absolute score accuracy.

5. A dimension-wise difficulty analysis revealing which narrative aspects LLMs can reliably assess.

1.6 Novelty of the Work

The novelty lies in jointly analyzing prompt sensitivity, model variability, and human alignment within a unified evaluation framework. We further highlight calibration effects, showing that high correlation does not necessarily imply accurate absolute scoring.

1.7 Paper Organization

The remainder of this paper is structured as follows. Section 2 reviews related work in automatic story evaluation and LLM-as-a-judge systems. Section 3 describes the dataset, models, and methodology. Section 4 presents experimental results. Section 5 discusses implications and limitations. Section 6 concludes the paper.

2 Related Work

2.1 Automatic Evaluation of Generated Text

The earliest widely adopted automatic metrics for generated text are based on n-gram overlap with a reference. Papineni et al. (2002) proposed BLEU for machine translation, computing modified n-gram precision between a candidate sentence and one or more references. They reported that BLEU correlates highly with human judgments at the corpus level across four language pairs, and BLEU has since become the default reference-based metric for many generation tasks. Lin (2004) introduced ROUGE for summarisation, with variants based on n-gram recall (ROUGE-N), longest common subsequence (ROUGE-L), and skip-bigrams (ROUGE-S); ROUGE remains the standard summarisation metric.

Both BLEU and ROUGE depend on surface form, and several authors have shown that this makes them unreliable for tasks in which many distinct outputs are simultaneously valid. Zhang et al. (2020) addressed this with BERTScore, which computes token-level similarity using contextual embeddings from BERT rather than exact n-gram matches. Evaluating on 363 machine-translation and image-captioning systems, they reported higher correlation with human judgments than BLEU, ROUGE, and METEOR, and stronger robustness to paraphrase.

These metrics were designed for tasks with a small space of acceptable outputs. Open-ended generation—and story generation in particular—admits many valid outputs that share little surface form with any reference, which is the gap the next line of work addresses.

2.2 Multi-Criterion Evaluation of Story Generation

Chhun et al. (2022) introduced HANNA (Human-ANnotated NArratives), a benchmark of 1,056 stories produced by ten automatic story-generation systems from prompts in the WritingPrompts dataset. Three annotators rated each story on six criteria drawn from the social-science literature on narrative: relevance, coherence, empathy, surprise, engagement, and complexity. Using these annotations, the authors meta-evaluated 72 automatic metrics—both reference-based and reference-free, string-based, embedding-based, and model-based—and found that no single metric correlates strongly with human judgment across all six criteria. They reported in particular that BLEU and ROUGE are among the weakest performers for evaluating creative narrative quality. We adopt HANNA as our evaluation benchmark for exactly this reason: it provides per-criterion human ratings that allow LLM-generated scores to be compared against humans on each dimension independently.

2.3 LLM-as-a-Judge

A more recent line of work uses LLMs themselves as automatic evaluators of generated text. Chiang and Lee (2023) showed that LLMs given the same instructions and rubrics as human annotators produce stable, reproducible ratings consistent with expert humans on open-ended story generation. Building on this, Liu et al. (2023) proposed G-Eval, which combines task-specific prompts with auto-generated chain-of-thought evaluation steps; using GPT-4, it reaches a Spearman correlation of 0.514 with humans on the SummEval summarisation benchmark, substantially above reference-based metrics. Zheng et al. (2023) provided a broader analysis using MT-Bench and Chatbot Arena, reporting roughly 80% agreement between GPT-4 and human preferences but identifying systematic failure modes including position, verbosity, and self-enhancement biases. Together, these findings establish LLM-as-a-judge as viable for our setting while motivating our use of multiple model families and explicit calibration analysis.

2.4 Prompting Strategies for LLM Evaluators

The behaviour of an LLM judge depends heavily on how it is prompted. Kojima et al. (2022) showed that simply appending the phrase "Let's think step by step" to a zero-shot prompt — which they call Zero-shot Chain-of-Thought — raises accuracy on MultiArith from 17.7% to 78.7% and produces large gains on other reasoning benchmarks, all without any task-specific examples. Their result established that small changes to prompt wording can have outsized effects on LLM behaviour. Liu et al. (2023) applied a related idea to evaluation specifically, generating chain-of-thought evaluation steps automatically before scoring, and reported improved alignment with human ratings compared to plain zero-shot evaluation prompts.

Findings on role and persona prompting are more mixed. Zheng et al. (2024) conducted a systematic evaluation of 162 personas across four LLM families on 2,410 factual questions. They reported that adding a persona to the system prompt does *not* improve performance overall: effects are small, are sometimes negative, and depend heavily on which specific persona is chosen. They also found that automatically identifying the best persona for a given question is no better than random selection. While their study targets factual reasoning rather than story evaluation, their results suggest that more elaborate or expert-sounding personas are not reliably better than simpler prompts — a hypothesis we test directly in the context of story evaluation by comparing role-based prompts of increasing specificity.

3 Method

This section describes the dataset, models, prompting strategies, evaluation metrics, and implementation details used to investigate whether LLMs can reliably evaluate story quality in alignment with human judgment.

3.1 Dataset

We use the **HANNA (Human-ANnotated NArratives)** dataset (Chhun et al., 2022), which contains 1,056 generated stories from the WritingPrompts dataset, each evaluated by three human annotators across six narrative quality dimensions:

- **Relevance**: How well the story matches the given prompt
- **Coherence**: Logical consistency and structural integrity
- **Empathy**: Emotional depth and ability to evoke feelings
- **Surprise**: Degree of unexpectedness or originality
- **Engagement**: How interesting and captivating the story is
- **Complexity**: Richness, depth, and narrative sophistication

Each dimension is scored on a 1–5 integer scale, where 1 = very poor and 5 = excellent. The three human annotations per story were averaged to produce a single ground-truth score for each dimension, which we treat as the reference standard for evaluating LLM performance.

HANNA was chosen because: (1) it provides fine-grained, multi-dimensional human annotations specifically designed for narrative evaluation, (2) it covers stories generated by diverse automatic systems, ensuring variety in quality and style, and (3) no single reference-based metric (BLEU, ROUGE, BERTScore) correlates strongly with human judgment across all six dimensions (Chhun et al., 2022), making human-annotated scores the most reliable ground truth.

Data preparation: We merged human annotations with LLM-generated scores on a per-story basis. Each row in the final dataset contains: `story_id`, human scores (`human_relevance`, `human_coherence`, etc.), LLM scores (`avg_relevance`, `avg_coherence`, etc.), model identifier (`model`), model version (`model_version`), and prompting technique (`technique`). Stories with missing LLM scores were excluded (2 stories, 20 rows total), yielding a final dataset of 10,540 evaluations across 1,054 stories.

3.2 LLM Evaluators

We evaluated two model families:

GPT-4.1 Mini (OpenAI): Tested across all five prompting techniques. This model was selected for its strong instruction-following capabilities and wide adoption in LLM-as-a-judge applications (Zheng et al., 2023).

Claude (Anthropic): Due to API cost constraints, we used different Claude variants across techniques:

- **Claude Sonnet 4.5**: Used for Role 1 (Human Annotator), Role 2 (Story Expert), Role 3 (Writing Professor), and Zero-Shot prompting
- **Claude Haiku 4.5**: Used for Instruction-Based prompting
- **Claude 3 Haiku**: Used for Role 1 prompting in version validation analysis only (not included in main results)

This resulted in **9 model-technique combinations** in the main analysis: GPT-4.1 Mini (5 techniques: Role 1, Role 2, Role 3, Instruction-Based, Zero-Shot) + Claude Sonnet 4.5 (4 techniques: Role 1, Role 2, Role 3, Zero-Shot) + Claude Haiku 4.5 (1 technique: Instruction-Based).

Version confound: Because the Instruction-Based condition uses Claude Haiku 4.5 while all other Claude techniques use Sonnet 4.5, comparisons between Claude Instruction-Based and other Claude techniques involve a model version difference. We acknowledge this limitation but do not directly validate it, as our validation analysis (Section 4.4) compares only Claude 3 Haiku and Sonnet 4.5.

Validation study: To assess whether model version affects evaluation quality, we conducted a validation comparing Claude 3 Haiku and Claude Sonnet 4.5 on identical stories using the Role 1 prompt (Section 4.4). This analysis found moderate cross-version correlation (Spearman $\rho = 0.557$), indicating the versions produce measurably different evaluations. We therefore use Sonnet 4.5 for Role 1 in the main analysis to maintain consistency with other non-Instruction Claude techniques.

3.3 Prompting Techniques

We designed five prompting strategies to test how prompt framing affects evaluation quality. These five strategies span three design axes: persona specificity (Role 1 $\rightarrow$ Role 2 $\rightarrow$ Role 3, in increasing elaboration), procedural scaffolding (Instruction-Based), and a no-guidance baseline (Zero-Shot). This design allows us to test whether more elaborate prompts produce better evaluations and whether procedural structure can substitute for persona framing.

1. Role 1 (Human Annotator): The model is instructed to act as a "trained human annotator working on a research dataset for narrative evaluation" following "strict guidelines used in academic benchmarks." The prompt emphasizes objectivity, consistency, and critical evaluation. This represents a simple, task-aligned persona without excessive elaboration.

2. Role 2 (Story Expert): The model is framed as someone "known for your ability to interpret stories by focusing on their underlying structure, emotional resonance, and narrative flow." This prompt adds domain expertise but remains concise, positioning the model as a skilled reader rather than a formal annotator.

3. Role 3 (Creative Writing Professor): The model is given an elaborate persona: "a published author and creative writing professor with over 20 years of experience teaching narrative craft and evaluating student work at the graduate level." This prompt includes detailed expertise claims, explicit evaluation philosophy ("rigor, nuance, and craft-informed judgment"), and expanded rubric descriptions. It represents the most complex prompting strategy tested.

4. Instruction-Based: A structured, step-by-step prompt that explicitly directs the model to: (1) read the story, (2) review the evaluation metrics, (3) evaluate based only on provided definitions, (4) assign scores, (5) provide justifications, and (6) format output as JSON. This approach emphasizes procedural clarity over persona.

5. Zero-Shot: Minimal prompting with no persona, no procedural steps, and only basic task framing: "Evaluate the following story using a structured rating system." This serves as the baseline for comparison.

All five prompts enforce the same output schema: a JSON object containing six integer scores (1–5) and corresponding justifications (1–2 sentences each). This standardization ensures that differences in performance reflect prompt framing rather than output structure variability.

Why we fixed the output schema: Allowing free-form outputs would introduce uncontrolled variance in response format, making comparisons between techniques unreliable. By constraining all prompts to produce identical JSON structures, we isolate the effect of prompt content (persona, instructions, framing) on evaluation quality.

Multiple evaluations per story: Each story was evaluated **three times** per model-technique combination to reduce stochastic variation in LLM outputs. The three scores per dimension were averaged to produce a single final score (`avg_relevance`, `avg_coherence`, etc.), which was then compared against the averaged human annotation for that story. Per-run variance was not analyzed and is left for future work.

3.4 Evaluation Metrics

We assess LLM performance using three categories of metrics: error-based, agreement-based, and correlation-based.

3.4.1 Error Metrics

Mean Absolute Error (MAE):

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i^{human} - y_i^{LLM}| \tag{1}$$

where y_i^{human} is the averaged human score and y_i^{LLM} is the averaged LLM score for story i. Lower values indicate better alignment.

Root Mean Square Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i^{human} - y_i^{LLM})^2} \tag{2}$$

RMSE penalizes large errors more heavily than MAE, revealing whether misalignments are systematic or driven by outliers.

3.4.2 Agreement Metrics

Exact-Match Accuracy: The percentage of cases where the rounded LLM score exactly matches the rounded human score. This treats evaluation as a 5-class classification problem.

Off-by-One Accuracy: The percentage of cases where the LLM score is within ± 1 of the human score. This more lenient metric captures whether the model is "directionally correct" even when not perfectly calibrated.

Cohen's Kappa (κ):

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{3}$$

where p_o is observed agreement and p_e is expected agreement by chance. Kappa adjusts for chance agreement, with values < 0.2 indicating poor agreement, 0.2–0.4 fair, 0.4–0.6 moderate, 0.6–0.8 substantial, and > 0.8 near-perfect.

Confusion Matrix: We visualize prediction patterns for the best and worst model-technique combinations using confusion matrices, which reveal systematic over- or under-prediction tendencies.

3.4.3 Correlation Metrics

Spearman's Rank Correlation (ρ): Measures monotonic association between human and LLM rankings. A non-parametric measure robust to outliers and suitable for ordinal data.

Pearson's Correlation (r): Measures linear relationship between human and LLM scores. Sensitive to the magnitude of differences, not just ranking order.

Both correlations range from -1 to 1, with values near 0 indicating no relationship and values near ± 1 indicating strong (anti-)correlation.

Composite Score: To facilitate ranking across configurations, we compute a single composite metric per model-technique combination by combining normalized MAE (inverted), exact-match accuracy, and Spearman correlation. Each metric is min-max normalized to [0,1], MAE is inverted so higher values indicate better performance, and the three normalized values are averaged. Higher composite scores indicate better overall alignment with human judgment. This composite is used only for ranking in comparative analyses; primary conclusions are drawn from the individual metrics.

3.5 Implementation Details

Platform: All experiments were conducted in Python 3.10 using Google Colab with standard scientific computing libraries (NumPy, pandas, SciPy, scikit-learn, Matplotlib, Seaborn).

Data processing: LLM evaluation results were stored in separate CSV files per model-technique combination. These files were merged on `story_id` with the human annotations to produce the unified analysis dataset. Missing values were handled by removing entire stories (not individual rows) to ensure balanced representation across all 10 configurations.

Reproducibility: All code, prompts, and aggregated results (excluding the full HANNA dataset due to licensing) are available in a GitHub repository. The analysis notebook is fully executable and produces all tables and figures reported in this paper.

Column naming convention:

- `human_*`: Human ground-truth scores (e.g., `human_relevance`)
- `avg_*`: LLM scores averaged across 3 evaluations (e.g., `avg_relevance`)
- `model`: Model family identifier ("GPT" or "Claude")
- `model_version`: Specific model variant (e.g., "Claude Sonnet 4.5")
- `technique`: Prompting strategy (e.g., "role_1", "instruction")

This unified schema enabled systematic comparisons across all model-technique-metric combinations in the subsequent analysis.

4 Results

We evaluated ten model-technique configurations across 1,054 stories and six narrative dimensions, yielding 10,540 individual score comparisons against human annotations. We report results across five complementary analyses: prompting technique comparison (section 4.1), head-to-head model comparison (section 4.2), dimension-level difficulty (section 4.3), Claude version validation (section 4.4), and calibration analysis (section 4.5).

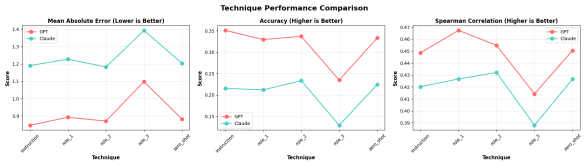

Figure 1: Composite score by technique for GPT-4.1 Mini and Claude. Role 3 fails catastrophically for both models. GPT's best technique (Role 1, 0.940) exceeds Claude's best (Role 2, 0.471) by a factor of two, indicating a model-level capability gap that prompt engineering cannot bridge.

4.1 Prompting Technique Comparison

Role 1 is the strongest technique for GPT; Role 2 is the strongest for Claude. Table 1 reports MAE, RMSE, exact-match accuracy, off by one accuracy, and Spearman correlation averaged across all six narrative dimensions for each model technique combination, together with a composite score that normalizes and equally weights the three primary metrics (lower MAE is better; higher accuracy and correlation are better).

For GPT-4.1 Mini, Role 1, Instruction-Based, and Role 2 form a high-performing cluster with composite scores of 0.940, 0.921, and 0.911 respectively, while Zero-Shot remains competitive at 0.881. Role 3 collapses to a composite score of 0.449—a 52% drop from the best technique. For Claude, all four reasonable techniques cluster tightly between 0.388 and 0.471, suggesting a model level ceiling rather than strong prompt sensitivity. Claude Role 3 scores 0.000, the lowest of any configuration tested.

Notably, off by one accuracy reveals that both models are closer to human scores than exact match figures suggest. GPT Instruction-Based achieves 71.5% off byone accuracy despite only 35.1% exact-match accuracy, and Claude Role 2 reaches 51.9% off-by-one versus 23.3% exact match. This two-to-one ratio indicates a systematic calibration gap: models correctly identify the approximate quality range but struggle to pin down the precise integer on the 1–5 scale.

Model	Technique	MAE↓	RMSE↓	Acc (%)↑	$\text{Acc}_{\pm 1}$ (%)↑	Spearman↑
GPT-4.1 Mini	Role 1 (Human Annotator)	0.892	1.080	33.0	68.6	0.467
GPT-4.1 Mini	Instruction-Based	0.846	1.032	35.1	71.5	0.449
GPT-4.1 Mini	Role 2 (Story Expert)	0.870	1.055	33.7	69.8	0.455
GPT-4.1 Mini	Zero-Shot	0.882	1.068	33.4	69.2	0.451
GPT-4.1 Mini	Role 3 (Writing Professor)	1.098	1.275	23.5	56.7	0.414
Claude	Role 2 (Story Expert)	1.182	1.365	23.3	51.9	0.432
Claude	Zero-Shot	1.204	1.385	22.4	50.7	0.427
Claude	Instruction-Based	1.190	1.374	21.5	52.2	0.420
Claude	Role 1 (Human Annotator)	1.228	1.408	21.2	49.1	0.427
Claude	Role 3 (Writing Professor)	1.394	1.562	12.9	40.2	0.388

Table 1: Performance of each model technique combination averaged across all six narrative dimensions (Relevance, Coherence, Empathy, Surprise, Engagement, Complexity). Acc = exact-match accuracy; $\text{Acc}_{\pm 1}$ = off-by-one accuracy (prediction within ± 1 of human score). GPT Role 1 achieves the highest composite score (0.940); Claude Role 3 the lowest (0.000). Role 3 degrades both models substantially relative to all other techniques.

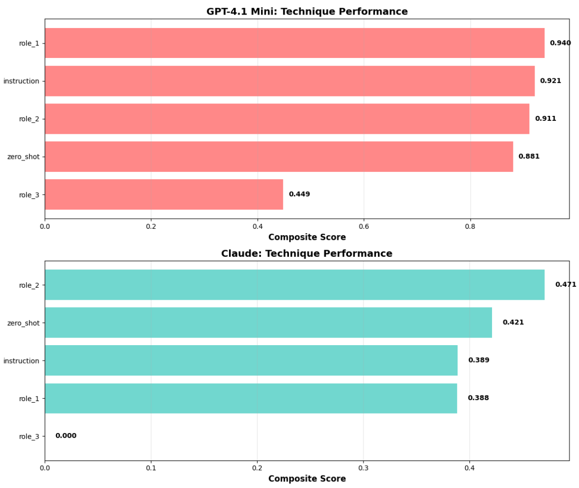

Figure 2: Ranked composite scores for all model technique combinations. GPT techniques occupy the top four positions; Claude techniques are confined to the lower half of the ranking.

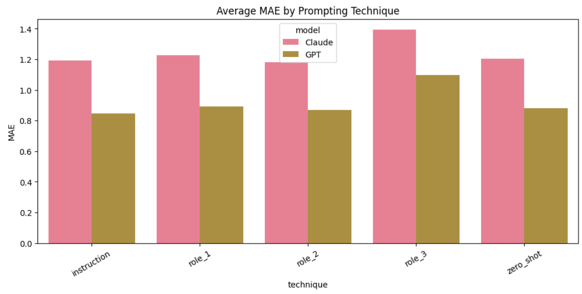

Figure 3: Average MAE by prompting technique for GPT-4.1 Mini and Claude. Role 3 produces the highest error for both models. GPT maintains a consistent ~0.3 point MAE advantage over Claude regardless of technique.

4.2 GPT-4.1 Mini vs. Claude

GPT-4.1 Mini outperforms Claude on every metric across all shared techniques. Table 2 summarises the two models when evaluated on the four techniques available to both (Role 1, 2, 3, Zero-Shot).

Model	MAE↓	Acc (%)↑	Spearman↑
GPT-4.1 Mini	0.936 ± 0.290	30.9 ± 11.6	0.433 ± 0.057
Claude	1.252 ± 0.390	20.0 ± 12.5	0.405 ± 0.051

Table 2: Overall model comparison across four shared techniques (Role 1, 2, 3, Zero-Shot). Values are mean $\pm$ standard deviation across all six narrative dimensions. GPT-4.1 Mini achieves 25% lower MAE, 55% higher exact-match accuracy, and 7% higher Spearman correlation than Claude. GPT wins all five head-to-head technique comparisons (including Instruction-Based).

GPT-4.1 Mini achieves a mean MAE of 0.936 versus 1.252 for Claude—a gap of 0.316 points on the 1–5 scale. Exact-match accuracy is 30.9% for GPT against 20.0% for Claude, a relative improvement of 55%. Spearman correlation shows a smaller but consistent advantage (0.433 vs. 0.405). In a five-way head-to-head comparison covering all shared techniques, GPT wins on MAE in every case (5–0).

The accuracy gap is larger than the correlation gap, which is theoretically informative: both models preserve relative story rankings with moderate fidelity ($\rho \approx 0.40$–0.47) but GPT is substantially more accurate at producing the correct absolute score category. Claude shows higher MAE variance (0.390 vs. 0.290), indicating less consistent performance across the six narrative dimensions.

4.3 Dimension-Level Analysis

Coherence is universally the hardest dimension; Surprise is universally the easiest. Table 3 reports MAE and RMSE for each narrative dimension averaged across all techniques within each model.

Coherence produces the highest error for

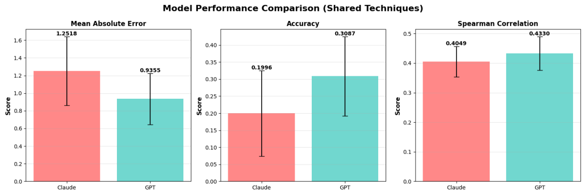

Figure 4: Head-to-head model comparison on MAE (lower is better), exact-match accuracy (higher is better), and Spearman correlation (higher is better). Error bars denote standard deviation across the six narrative dimensions. GPT-4.1 Mini leads on all three metrics.

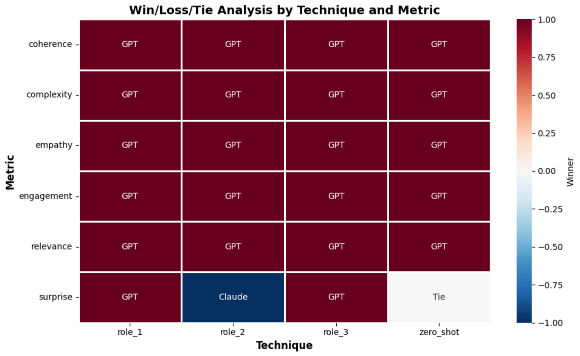

Figure 5: Win/loss heatmap showing which model achieves lower MAE for each technique-dimension combination. GPT-4.1 Mini wins the majority of cells across all techniques and dimensions.

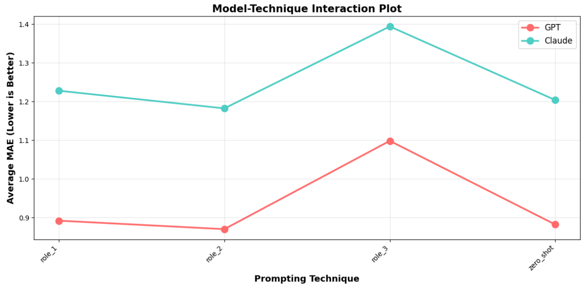

Figure 6: Model × technique interaction plot. The performance gap between GPT and Claude is consistent across all techniques, confirming that model choice has a larger effect than prompt choice.

2*Dimension	GPT		Claude	
	MAE	**RMSE**	**MAE**	**RMSE**
Coherence	1.512	1.660	1.929	2.053
Engagement	0.945	1.145	1.464	1.631
Relevance	0.792	0.994	1.269	1.488
Complexity	0.882	1.071	1.112	1.302
Empathy	0.726	0.913	0.944	1.126
Surprise	0.650	0.831	0.718	0.914
Average	**0.918**	**1.069**	**1.239**	**1.419**

Table 3: MAE and RMSE per narrative dimension, averaged across all techniques within each model. Coherence is the hardest dimension for both models; Surprise is the easiest. No technique or model achieves MAE < 1.5 on Coherence.

both models (GPT: MAE = 1.512; Claude: MAE = 1.929), with an average across both models of 1.720. No single technique or model achieves MAE below 1.5 on this dimension. Surprise is the easiest dimension for both models (GPT: 0.650; Claude: 0.718; average: 0.684), followed by Empathy (GPT: 0.726; Claude: 0.944; average: 0.835).

The pattern separates into two groups. Dimensions with objective, locally verifiable properties Surprise and Empathy are reliably assessed by both models. Dimensions requiring holistic comprehension of the entire narrative Coherence, Engagement, and Relevance produce substantially higher errors, particularly for Claude. This suggests that LLM evaluators can detect surface-level narrative signals but struggle with the integrative reasoning needed to assess structural narrative quality.

4.4 Claude Version Validation

Claude Haiku 3 and Sonnet 4.5 show only moderate agreement on Role 1, confirming a version confound in the dataset. Because API cost constraints required using Claude 3 Haiku for Role 1 and Claude Sonnet 4.5 for Roles 2, 3, and Zero-Shot, we tested whether the two versions produce comparable scores on the same stories. Table 4 reports per-dimension Spearman correlations between the two versions evaluated on the identical Role 1 prompt and story set (n = 1,050 matched pairs).

Dimension	**Spearman ρ**
Coherence	0.645
Engagement	0.570
Complexity	0.542
Empathy	0.544
Relevance	0.521
Surprise	0.519
Average	**0.557**

Table 4: Spearman correlation between Claude 3 Haiku and Claude Sonnet 4.5 scores on the same stories under Role 1 prompting (n = 1,050; all $p < 0.0001$). Average $\rho = 0.557$ indicates moderate agreement. The two versions are not interchangeable, introducing a measurable confound when comparing Claude Role 1 against other Claude techniques evaluated with Sonnet 4.5.

Average cross-version correlation is $\rho = 0.557$, which falls in the moderate range. Coherence shows the highest agreement ($\rho = 0.645$), while Surprise and Relevance show the weakest ($\rho \approx 0.52$). All correlations are statistically significant ($p < 0.0001$), but practical agreement is far from

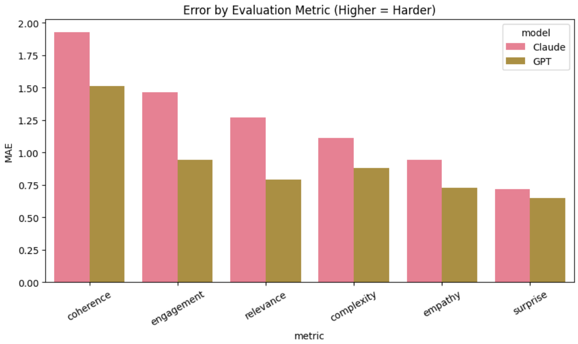

Figure 7: Average MAE per narrative dimension for GPT-4.1 Mini and Claude. Coherence is the hardest dimension for both models; Surprise and Empathy are the easiest.

the $\rho > 0.8$ that would be required to treat the versions as interchangeable evaluators. As a result, observed differences between Claude Role 1 and other Claude techniques must be interpreted with caution: they may reflect model version effects as much as prompt effects.

4.5 Calibration and Agreement

Both models rank stories with moderate fidelity but fail to produce calibrated absolute scores. Cohen's Kappa across all configurations ranges from -0.002 Claude Role 3, effectively random) to 0.107 (GPT Instruction-Based), indicating near-zero inter-rater agreement between LLM and human scores in absolute terms. Despite this, Spearman correlations consistently range from 0.39 to 0.47, showing that relative orderings of story quality are partially preserved.

This divergence between ranking ability ($\rho \approx 0.43$) and absolute agreement ($\kappa < 0.11$) constitutes a calibration gap with practical consequences: LLM scores cannot be used as direct substitutes for human ratings without post-hoc normalisation, but they may remain useful for ranking or filtering at scale.

The off-by-one analysis further clarifies the nature of the errors. GPT Instruction-Based achieves 71.5% off-by-one accuracy versus 35.1% exact-match accuracy a 2.0× improvement when the tolerance window is widened by one point. For Claude Role 2, the corresponding figures are 51.9% and 23.3% (a 2.2× improvement). This pattern holds across all configurations and confirms that most errors are small adjacent-category misclassifications rather than large systematic biases.

5 Discussion

5.1 Why the Best Techniques Won and the Worst Failed

Instruction-Based and Role 1 emerged as the strongest prompting strategies because they balanced structure with simplicity. Instruction-Based provided explicit procedural steps—read, evaluate, assign scores, format as JSON—creating a clear cognitive pathway that reduced arbitrary outputs. Role 1 framed evaluation as objective annotation work with "strict guidelines used in academic benchmarks," which aligned naturally with the structured output format. Both emphasized clarity and consistency without cognitive overhead.

Role 2 performed comparably well despite its more conversational framing. The "story expert" persona added domain-specific language ("underlying structure," "emotional resonance") without the excessive elaboration that degraded Role 3. Zero-Shot demonstrated that both models possess reasonable baseline evaluation capabilities, though adding minimal structure (Role 1) or procedural guidance (Instruction-Based) consistently improved performance.

Role 3 failed catastrophically for both models. The elaborate "creative writing professor" persona—complete with 20 years of experience claims, graduate-level evaluation philosophy, and craft-specific terminology—introduced cognitive overhead that actively degraded performance. Rather than improving evaluation quality, the complexity caused models to prioritize mimicking expert discourse over producing accurate numerical scores. This resulted in systematic miscalibration, with GPT's composite score dropping 52% and Claude's collapsing entirely to 0.000.

This pattern strongly supports findings from Zheng et al. (2024), who reported that elaborate personas do not reliably improve LLM performance and often degrade it. Our results extend this finding to the evaluation domain: simpler, task-aligned prompts consistently outperform sophisticated personas across both model families and all narrative dimensions.

5.2 Model-Level Performance Differences

GPT-4.1 Mini outperformed Claude across every metric and technique combination. The 25% MAE advantage (0.936 vs 1.252) and 55% accuracy advantage (30.9% vs 20.0%) represent substantial, practically meaningful gaps. Critically, the per-

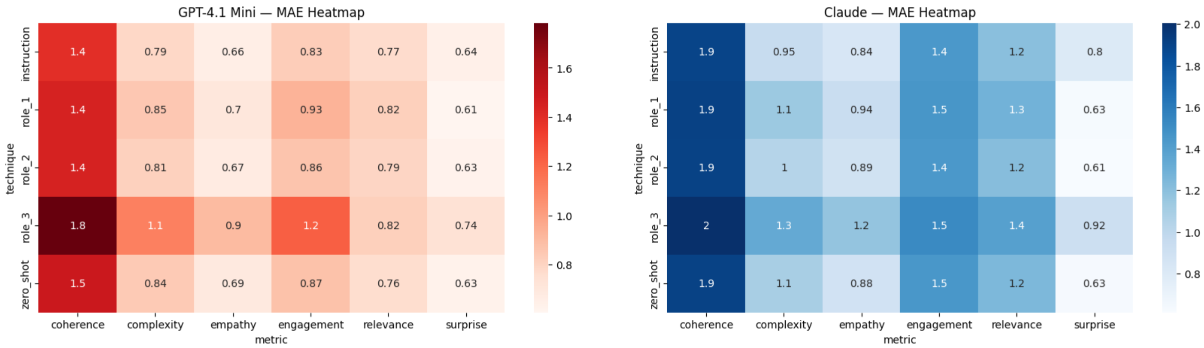

Figure 8: MAE heatmaps per technique × dimension for GPT-4.1 Mini (left) and Claude (right). Darker cells indicate higher error. Coherence is the darkest column for both models. Role 3 is the darkest row, confirming it degrades performance across all dimensions.

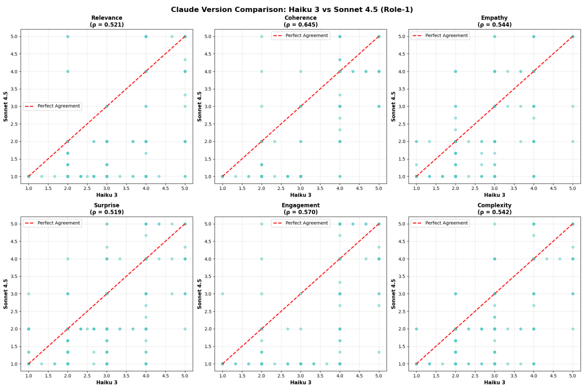

Figure 9: Scatter plots comparing Claude 3 Haiku vs Claude Sonnet 4.5 scores on the same stories under Role 1 prompting, one panel per narrative dimension. Points on the diagonal indicate perfect agreement. Substantial scatter across all six panels confirms the two versions are not interchangeable.

formance difference remained consistent across prompting techniques—no prompt design bridged the model capability gap.

The interaction analysis (section 4.2) confirmed this: parallel lines in the model-technique plot indicate that both models respond similarly to prompt changes. Role 3 degrades both; Instruction-Based and Role 1 improve both. The vertical gap between lines remains constant, demonstrating that model choice has a larger effect on evaluation quality than prompt choice.

This has direct implications for practitioners: **select the better model first, then optimize prompts second.** Investing effort into elaborate prompt engineering for a weaker model yields diminishing returns compared to switching to a stronger model with a simple prompt.

The divergence also reveals different calibration baselines. Claude assigned systematically lower scores across all dimensions, suggesting stricter internal evaluation standards or different anchor points for the 1–5 scale. This cross-model calibration difference complicates ensemble approaches and underscores the need for model-specific post-hoc normalization before scores can be interpreted as equivalent to human ratings.

5.3 The Ranking-vs-Calibration Gap

Both models demonstrated moderate ranking ability (Spearman $\rho \approx 0.40$–0.47) but poor absolute score calibration (Cohen's Kappa < 0.11). This divergence is the study's most important finding for practical deployment.

The models can order stories by quality—identifying which story is better in pairwise comparisons—but struggle to assign the correct integer score on the 1–5 scale. Off-by-one accuracy doubles exact-match accuracy (GPT: 71.5% vs 35.1%; Claude: 51.9% vs 23.3%), confirming that errors are primarily adjacent-category misclassifications rather than random noise or wild guessing.

This calibration gap has three practical consequences. First, **LLM scores cannot substitute directly for human ratings** without post-hoc normalization. Raw outputs will systematically misrepresent story quality in absolute terms. Second, **LLMs remain useful for ranking and filtering tasks at scale.** Applications requiring "top 10% of submissions" or "better than baseline" decisions can leverage the preserved relative orderings. Third, **evaluation protocols must be designed around model strengths.** Systems should rank or compare rather than score, or implement explicit calibration layers that map LLM outputs onto human-equivalent scales.

The gap also suggests a fundamental limitation:

Figure 10: Confusion matrices for the best-performing configuration (GPT-4.1 Mini + Instruction-Based, left) and the worst-performing configuration (Claude + Role 3, right) on the Relevance dimension. The best configuration shows scores distributed around the diagonal; the worst collapses to a near-constant prediction, confirming a qualitative failure mode rather than mere miscalibration.

current LLMs lack the precise numerical grounding humans develop through repeated calibration exercises. Humans internalize anchor examples ("this is a 3") and maintain consistency through reference to past judgments. LLMs generate scores through next-token prediction without access to stable internal anchors, leading to systematic drift even when relative orderings remain valid.

5.4 Dimension-Specific Difficulties

Coherence proved universally hardest to evaluate (average MAE 1.72), while Surprise was easiest (average MAE 0.68). This 2.5× difficulty gap reveals which narrative qualities LLMs can assess reliably and which remain challenging.

Surprise and Empathy (MAE 0.68 and 0.84 respectively) are easier to evaluate because they have concrete textual markers: plot twists, unexpected events, emotional language, and expressions of character depth. Models can identify these features by reading individual sentences or short passages without needing to understand the full story. For example, a sudden plot revelation or emotionally charged dialogue can be recognized locally.

Coherence, Engagement, and Relevance (MAE 1.72, 1.20, and 1.03) require **holistic integration**: tracking cause-effect across the entire story, maintaining awareness of the original prompt throughout evaluation, and assessing whether narrative momentum sustains reader interest from beginning to end. These demands exceed current LLM capabilities for integrative reasoning over long contexts.

Importantly, this difficulty pattern held across all five prompting techniques, indicating it is a **model-level limitation rather than a prompt-design failure.** No amount of persona elaboration or procedural scaffolding enabled models to assess coherence as reliably as they assessed surprise.

The finding complicates ground-truth reliability. Coherence and engagement are inherently more subjective than surprise—human annotators themselves may disagree on whether a story "flows logically" or "maintains interest." When averaged human ratings carry substantial inter-annotator variance, it becomes unclear whether low LLM-human correlation reflects model failure or genuine ambiguity in the construct being measured (Chhun et al., 2022).

5.5 Comparison with Prior LLM-as-Judge Work

These findings align with and extend prior research. The ranking-vs-calibration gap replicates results from Zheng et al. (2023) and Liu et al. (2023), who reported that LLMs preserve relative orderings better than absolute scores across multiple evaluation domains. Our contribution is demonstrating this pattern holds specifically for multi-dimensional narrative evaluation, where subjective dimensions (coherence, engagement) exacerbate the calibration problem.

The prompt sensitivity analysis extends Zheng et al. (2024) from factual reasoning to creative evaluation. Their finding—that elaborate personas often degrade performance—generalizes to our setting, with Role 3's catastrophic failure providing the most extreme confirmation. We add the novel observation that **procedural prompts (Instruction-Based) can match or exceed persona-based prompts (Role 1, Role 2)** without requiring domain-specific framing.

The self-preference bias concern raised by Panickssery et al. (2024) remains unaddressed by our design, as we did not test whether models favor stories generated by their own family. Future work should directly compare GPT evaluating GPT-generated vs Claude-generated stories to isolate this effect.

5.6 Limitations

Dataset scope: All stories are English-language, limiting generalizability to multilingual or low-resource settings. The HANNA benchmark covers only short-form fiction from WritingPrompts; results may differ for other genres (journalism, technical writing, poetry) or lengths (novels, flash fiction).

Model coverage: Only two model families were evaluated. Broader claims about "LLMs in general" require testing across Llama, Gemini, Mistral, and other architectures. The GPT-Claude gap observed here may not generalize to other pairwise comparisons.

Version confounds: Inconsistent Claude versioning (Haiku 3 for Role 1, Sonnet 4.5 for others, Haiku 4.5 for Instruction-Based) introduces confounds that complicate clean technique comparisons within the Claude family. While we validate the Haiku 3 vs Sonnet 4.5 difference in section 4.4, the Instruction-Based condition remains unvalidated. Future work should use a single model version across all techniques.

Fixed prompts: Prompts were designed based on existing literature but not iteratively optimized. Performance could potentially improve with more carefully tuned templates, particularly for Role 3, though the magnitude of its failure suggests fundamental issues beyond wording refinement.

Ground truth subjectivity: Human annotations are treated as ground truth despite inherent inter-annotator disagreement, particularly for dimensions like coherence and engagement. When humans themselves disagree, "alignment with human judgment" becomes ambiguous—which human? Averaged ratings obscure this variance.

Lack of self-preference bias testing: We did not control for whether stories were generated by the same model family being used for evaluation. This leaves open the question of whether observed performance reflects genuine evaluation capability or model-specific bias.

Lack of annotator metadata: The HANNA benchmark anonymizes human annotators to protect privacy, preventing analysis of whether individual annotator characteristics (evaluation style, expertise, personality) correlate with specific prompting techniques. We cannot determine if role-based prompts (e.g., "creative writing professor") align better with certain annotator profiles or if prompt effectiveness varies by evaluator archetype. Future work with non-anonymized annotator metadata could examine personality-prompt interactions and enable tailored prompt optimization for different human judgment styles.

5.7 Practical Recommendations

For practitioners deploying LLM-based story evaluation:

Choose the stronger model. Model selection has a larger effect than prompt engineering. GPT-4.1 Mini outperformed Claude across all configurations; investing in the better model yields more improvement than optimizing prompts for the weaker one.

Use simple, structured prompts. Instruction-Based and Role 1 match or exceed more elaborate alternatives. Avoid complex personas (Role 3) that introduce cognitive overhead without performance gains.

Design around ranking, not scoring. Leverage preserved relative orderings for filtering (top 10%), comparison (Story A vs Story B), or threshold decisions (above/below baseline). Avoid direct interpretation of raw scores as human-equivalent ratings.

Implement calibration layers. If absolute scores are required, apply post-hoc normalization mapping LLM outputs onto human-equivalent scales. This requires collecting paired LLM-human annotations on a calibration set.

Validate on your domain. Our results reflect short-form English fiction. Performance may differ for other genres, lengths, or languages. Pilot testing against human annotations on domain-specific samples is essential before deployment.

Treat LLMs as complements, not replacements. Current systems are best understood as scalable pre-filters that reduce human annotation load rather than full substitutes for expert human judgment.

6 Conclusion

As large language models become increasingly embedded in creative writing tools, content generation pipelines, and automated evaluation systems, understanding their reliability as judges of narrative quality is critical. This study addressed whether LLMs can evaluate story quality in alignment with human judgment, or whether calibration failures undermine their utility.

We conducted a comprehensive evaluation of GPT-4.1 Mini and Claude variants across five prompting strategies (Role 1, Role 2, Role 3, Instruction-Based, Zero-Shot) using the HANNA benchmark of 1,056 human-annotated stories. Over 10,000 individual evaluations were compared against human annotations across six narrative dimensions using error metrics (MAE, RMSE), agreement metrics (accuracy, Cohen's Kappa), and correlation metrics (Spearman, Pearson).

Three primary findings emerged. First, **simpler prompts outperform elaborate personas.** Instruction-Based and Role 1 achieved the strongest performance, while the complex Role 3 "creative writing professor" persona failed catastrophically for both models. This confirms that cognitive overhead from elaborate framing degrades rather than improves evaluation quality.

Second, **model choice matters more than prompt design.** GPT-4.1 Mini outperformed Claude by 25% on MAE and 55% on accuracy across all techniques. The performance gap remained constant regardless of prompting strategy, demonstrating that no amount of prompt engineering bridges fundamental model capability differences.

Third, **LLMs rank well but calibrate poorly.** Both models preserved relative story orderings with moderate fidelity (Spearman $\rho \approx 0.40$–0.47) but achieved near-zero absolute agreement with humans (Cohen's Kappa < 0.11). Off-by-one accuracy doubled exact-match accuracy, indicating systematic adjacent-category errors rather than random miscalibration. This ranking-vs-calibration gap has direct implications: LLM scores cannot substitute for human ratings without normalization, but remain useful for ranking and filtering tasks at scale.

These findings carry clear practical guidance. Practitioners should prioritize model selection over prompt optimization, use simple structured prompts rather than elaborate personas, design systems around ranking rather than absolute scoring, and implement calibration layers when scores are required. LLM-based evaluation is best understood as a scalable complement to human annotation rather than a replacement.

Future work should address four key limitations. First, **control model versioning.** Our use of different Claude variants introduced confounds; isolating technique effects requires consistent model versions across all conditions. Second, **expand to multilingual and multi-genre evaluation.** Our English-language short-story results may not generalize to other languages, domains, or narrative lengths. Third, **develop automatic calibration methods.** The ranking-vs-calibration gap could potentially be closed through learned mapping functions that normalize LLM outputs onto human-equivalent scales without manual annotation. Fourth, **test for self-preference bias.** Directly comparing model performance when evaluating same-family vs different-family generated stories would isolate whether observed alignment reflects genuine capability or systematic favoritism.

A promising research direction is **human-in-the-loop hybrid systems** where LLMs handle initial evaluation and ranking at scale, and humans review edge cases, resolve disagreements, and provide calibration signals. Such systems leverage LLM strengths (ranking, scalability) while mitigating weaknesses (miscalibration, dimension-specific failures) through targeted human oversight.

The central implication is clear: LLMs can meaningfully support narrative evaluation at scale, but only when deployed with awareness of their limitations. Prompt design, model selection, and calibration are not implementation details—they are prerequisites for responsible deployment. Until automatic calibration methods mature and model capabilities improve, LLM-based evaluation should augment rather than replace human judgment in high-stakes creative and evaluative contexts.

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